

TAMIL NADU STATE BOARD - STANDARD 12 CHEMISTRY VOLUME 2

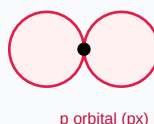
CHAPTER 11: HYDROXY COMPOUNDS AND ETHERS

PREMIUM ACADEMIC STUDY SUITE & EXAM GUIDE

Board:	Tamil Nadu State Board (TNSB)
Syllabus:	Samacheer Kalvi Textbook Curriculum aligned
Standard:	Class 12 English Medium
Subject:	Chemistry Volume 2
Chapter Name:	Hydroxy Compounds and Ethers

Learning Objectives:

- Comprehend core theoretical concepts and exact textbook terminology.
- Apply relevant formulas and proofs to solve high-scoring board problems.
- Interpret and sketch vector diagrams or graphical models representing equations.
- Build examination readiness for 1-mark, 2-mark, and 5-mark question sets.

Atomic Orbitals Wave Functions (s & p)

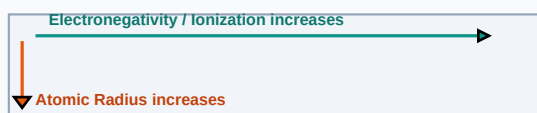
Core Concepts & Definitions

Textbook definitions are essential for scoring fully in the short-answer section of board assessments. Review the primary definitions below:

Term	Official Definition & Context
Lucas Test	A chemical test to differentiate primary, secondary, and tertiary alcohols based on turbidity speeds with ZnCl_2/HCl .
Williamson Synthesis	An organic reaction involving nucleophilic attack of an alkoxide ion on a primary alkyl halide to form ethers.
Phenol	An organic compound containing a hydroxyl group directly bonded to an aromatic benzene ring.

Syllabus Exam Tip: Memorize precise keywords. Examiners award maximum marks when textbook terms and correct symbols are correctly referenced.

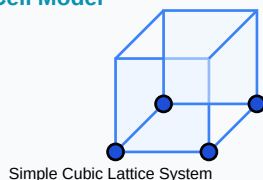
Periodic Properties Trends Direction Map



Detailed Explanation & Formulas

Here, we analyze the core laws and formulas. Examine the conceptual illustration below to visual the relation between variables:

Crystal Solid Lattice Unit Cell Model



Essential Formulas, Laws & Identities:

- Lucas turbidity speed: Tertiary (immediate) > Secondary (5 mins) > Primary (none at room temp)
- Williamson: $\text{R-O}^- \text{Na}^+ + \text{R}'\text{-X} \rightarrow \text{R-O-R}' + \text{NaX}$

Make sure to check the dimensional consistency of all physical quantities and verify that units are correctly stated in final solutions.

Worked Examples

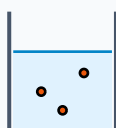
Review these standard board examination-style worked examples to learn proper step layout structures:

Example 1: Convert Phenol to Benzene.

Step-by-step Solution:

Heat Phenol ($\text{C}_6\text{H}_5\text{OH}$) with Zinc dust. Zinc reduces phenol, distilling pure Benzene (C_6H_6) and forming Zinc Oxide (ZnO).

Titration Beaker & Solute Precipitation



Reacting Mixture

HOTS (Higher Order Thinking Skills) Challenge

Challenge yourself with these complex, multi-concept problems designed to test deeper conceptual understanding and mathematical reasoning:

HOTS Challenge 1: Explain the mechanism of acid-catalyzed dehydration of Ethanol to form Ethene.

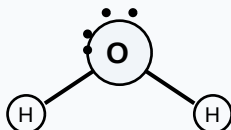
Analytical Solution Strategy:

Mechanism: (1) Protonation of alcohol: $\text{C}_2\text{H}_5\text{OH} + \text{H}^+ \rightarrow \text{C}_2\text{H}_5\text{OH}_2^+$.

(2) Carbocation formation (slow step): $\text{C}_2\text{H}_5\text{OH}_2^+ \rightarrow \text{CH}_3\text{-CH}_2^+ + \text{H}_2\text{O}$.

(3) Deprotonation (fast): $\text{CH}_3\text{-CH}_2^+ \rightarrow \text{CH}_2=\text{CH}_2 + \text{H}^+$, regenerating the acid catalyst.

Lewis Dot Structure (H₂O Bent Molecular Geometry)



Practice Worksheet

Test your skills by attempting these sample exam problems independently. Draw diagrams where appropriate:

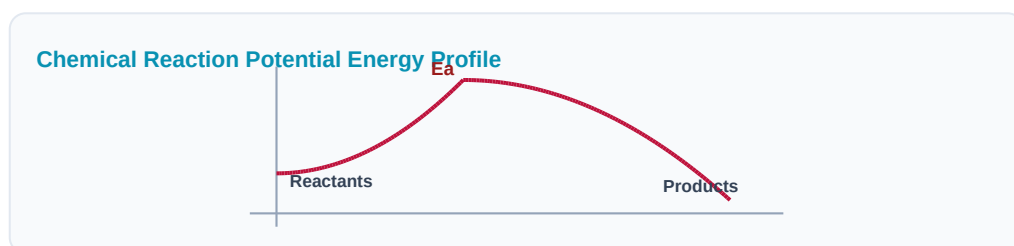
Q1 (1-Mark): How do you distinguish primary, secondary, and tertiary alcohols using the Victor Meyer test?

Q2 (2-Mark): Describe the preparation of phenol by Dow's Process.

Q3 (2-Mark): Explain Kolbe's Reaction of Phenol (salicylic acid formation).

Q4 (5-Mark): Write the equation for Reimer-Tiemann reaction.

Q5 (5-Mark): Describe the cleavage of ethers by Halogen acids (HI).



Real-World Applications & Connections

This chapter connects directly with modern industrial engineering, computation, and natural systems in numerous ways:

Application Case: Antiseptics

Phenol derivatives act as strong disinfectants in sanitizing hospital grids.

Application Case: Chemical Synthesis

Diethyl ether acts as an extraction solvent to synthesize organic intermediates.

Cross-Curricular Link: Concepts learned in this unit are heavily applied in other subjects like Physics (equations of motion, field vectors) and Data Analytics.

Aromatic Benzene Ring (C₆H₆) Resonance



Revision Summary & Strategy

Use this checklist to self-evaluate your exam preparation status before final tests:

Key Recall Tips:

- ✓ Always write down 'Given Parameters' first to ensure partial marks are locked in.
- ✓ Pay attention to signs (+ / -) when performing algebraic operations or force mappings.
- ✓ Check that you have written the correct units (e.g. Joules, Tesla, cm^3 , N) for the final numeric answer.
- ✓ Draw neat diagrams using a sharp pencil and ruler for geometry or circuit layouts.

Syllabus Checklist:

- ☐ Read and memorized core definitions on Page 2.
- ☐ Reviewed critical formulas and relations on Page 3.
- ☐ Solved worked examples on Page 4.
- ☐ Attempted worksheet problems on Page 6.

CONGRATULATIONS! YOU HAVE COMPLETED THIS SYLLABUS STUDY GUIDE!

Practice consistently, verify answers with standard textbook methods, and take the topic test in the Nexora Learning app to gauge progress.
Good luck!

Electrochemical Galvanic Cell System

